WORLDS WORKSHOP · IEEE/RSJ IROS 2026 · PITTSBURGH · SPOTLIGHT 1, PAPER 7

Modeling What Changes

Sparse, Residual World Models for Object-Centric Manipulation

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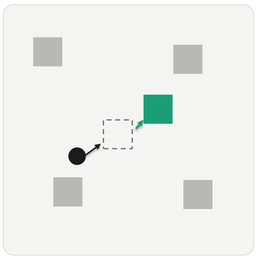
Code, data, checkpoints

and a live in-browser demo running the real models

github.com/ParamThakkar123/sparse_world_models

A push moves one object; the rest of the scene is identical. A per-object gate that copies unchanged objects exactly stops a world model from corrupting the static scene, at a tenth of the parameters.

1 Scenes are sparse in change



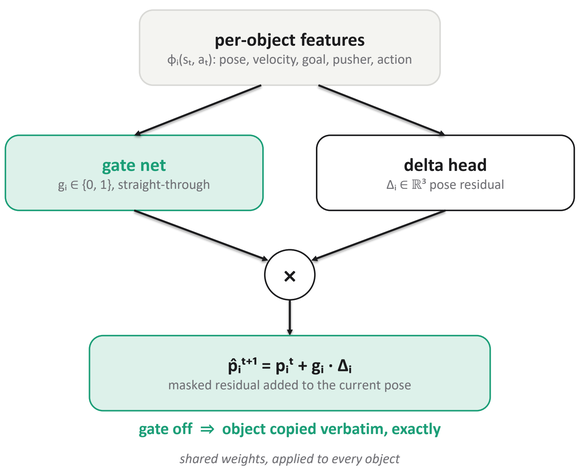
One push. One object moves.

Every other object is exactly where it was.

Dense world models still regress *every* pose, every step.

- **Capacity** is spent re-producing the static majority
- **Error injection:** a regressor perturbs objects it should copy; in rollout the error piles up on objects that never moved
- **No handle on "what":** no explicit belief about which objects will change

2 Gate what changes, copy the rest



gate off ⇒ object copied verbatim, exactly

shared weights, applied to every object

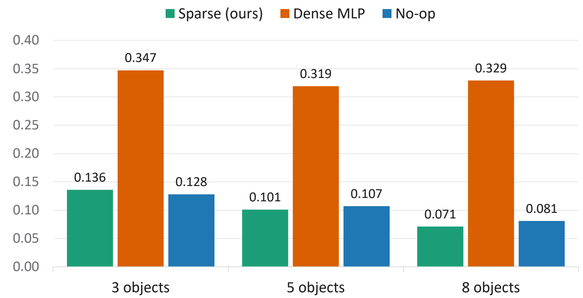
$$\mathcal{L} = \mathrm{BCE}(g, m^*) + \lambda\Delta \cdot m^* \cdot \|\Delta - \Delta^*\|^2 + \lambda s \cdot \bar{g}$$

- **Hard at inference:** un-gated objects get exactly zero update
- **Count-agnostic:** shared heads, nothing sized to N; train on 3 objects, run on 8
- **Tiny:** 2-layer width-128 heads, under 0.1M parameters

3 Controlled tabletop pushing

- **MuJoCo tabletop,** N ∈ {3, 5, 8} boxes, scripted pusher; structured state, not pixels
- **Baselines:** **Dense MLP** regresses all poses; **No-op** predicts no change, the floor to beat
- **Metric:** per-object pose L2, split into changed vs. unchanged objects; 3 seeds

4 Lower error, a tenth of the parameters

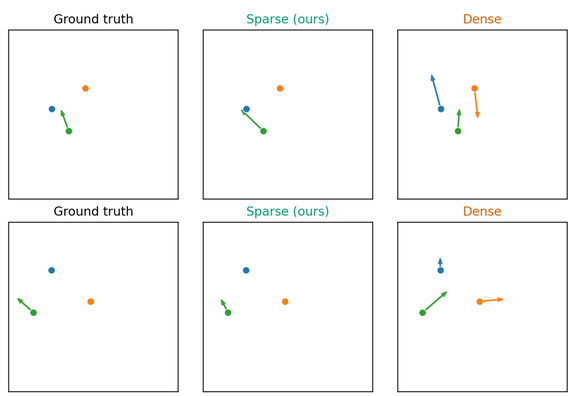


Overall per-object pose L2, held-out, lower is better (Table I).

2.5–4.6× lower L2 than dense

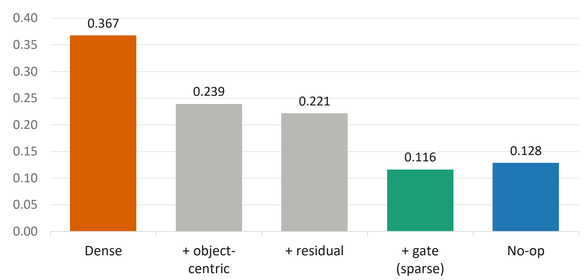
8.6–11.1× fewer parameters

F1 0.80–0.87 change detection; dense is degenerate



Held-out scenes. **Sparse moves only the pushed object; dense hallucinates motion on all three.**

5 Where the win comes from



Capacity-matched ladder, one ingredient per rung (Table II, 3 objects).

Unchanged-object L2: 0.211 → 0.111 → 0.086 → **0.001**

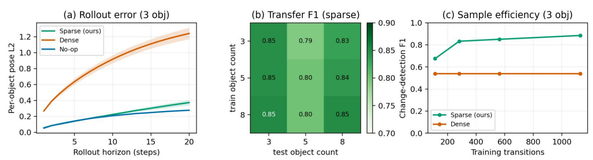
The gate is the largest single step, and the only one below no-op.

Changed-object L2 with a perfect (oracle) gate

N	predicted gate	oracle gate	no-op
3	0.312	0.314	0.348
5	0.426	0.432	0.446
8	0.466	0.474	0.470

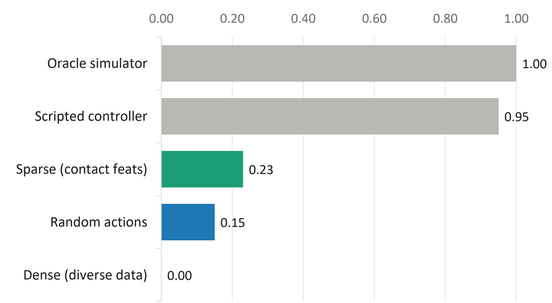
A perfect gate barely helps. The win is detecting what changed and not corrupting the rest. On the movers themselves the model is at no-op quality: delta regression is the bottleneck.

6 A world model: rollout, transfer, data



- **(a) Rollout:** 3.4–6.4× less 20-step error than dense, still above no-op
- **(b) Transfer:** one checkpoint, any N; 99.4% of F1 kept off-count. Dense cannot run off-count
- **(c) Data:** ~90% of full-data F1 from 25% of the data

7 Planning: beats dense, not yet random



Push-to-goal success, same CEM planner, only the model swapped; 3 seeds × 20 episodes (Table IV).

- **Round 1:** both learned models score 0.00; the planner queries off-distribution states
- **Round 2:** contact-aware features + diverse data: sparse 0.23 ± 0.06, dense 0.00 at every seed. Margin over random (0.15) is not significant

8 What it is, and is not

IT IS

a change detector that avoids error injection

far better than dense one-step, in rollout and in planning

count-invariant, with an inspectable mask

IT IS NOT

an accurate predictor of the movers

better than no-op in rollout

significantly better than random at planning, or faster in wall-clock

9 Takeaways

- **Model what changes:** a gate + masked residual is a one-line bias
- **Decompose the metric:** always report no-op and the changed / unchanged split
- **The delta head is the lever:** detection is solved, contact-delta regression is not